

SMART SHOPPING TROLLEY

A PROJECT REPORT

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NOVEMBER 2025

DECLARATION

We affirm that the Project Report titled **SMART SHOPPING TROLLEY** being submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In today's fast-paced retail environment, traditional shopping methods are often slow, inefficient, and inconvenient. Customers spend significant time standing in long billing queues and waiting for manual item scanning, especially during peak hours. These delays not only reduce customer satisfaction but also increase the workload on cashiers and staff. Moreover, shoppers often forget to buy necessary products or face uncertainty about the total bill while shopping. Though technologies such as RFID-based smart trolleys and self-checkout systems have been introduced to automate the process, they remain expensive and complex to implement in small and medium-sized stores due to high hardware and maintenance costs. To overcome these limitations, the Smart Shopping Trolley system is proposed — a low-cost, user-friendly, and efficient alternative that integrates embedded systems, cloud technology, and a smartphone-based interface. The aim is to make shopping faster, smarter, and more secure by enabling customers to handle billing and payments directly through their own mobile devices, without depending on the store's infrastructure.

The proposed system combines an ESP32 microcontroller, a load cell with HX711 amplifier, and a Firebase Realtime Database connected through Wi-Fi. During shopping, the customer uses a mobile application to scan the product barcode with the phone's camera. The app instantly retrieves product information from a database and updates the total bill in real time. Simultaneously, the load cell measures the product's weight to verify that the scanned item matches what is placed in the trolley, thereby preventing theft or mismatch. The ESP32 transmits this weight data to the Firebase cloud, ensuring live synchronization between the hardware and mobile application. Once the customer finishes shopping, payment can be made directly through the integrated mobile app, eliminating the need for physical checkout counters. The system was simulated using Wokwi in Visual Studio Code, and practical testing confirmed smooth data flow between the ESP32, cloud, and mobile app. The results demonstrate that the Smart Shopping Trolley effectively reduces billing time, minimizes manual work, and ensures accurate, real-time billing. By integrating affordability, automation, and security, the system provides a scalable solution suitable for both small retailers and large supermarkets. The successful implementation highlights how embedded systems and IoT can transform traditional shopping into a faster, smarter, and customer-centered experience.

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CHAPTER – 1

INTRODUCTION

1.1 INTRODUCTION

In the modern era, supermarkets and shopping malls have become an integral part of people's lives, serving as one-stop destinations for purchasing daily essentials and lifestyle products. However, despite technological advancements in retail management, traditional shopping methods remain time-consuming and inefficient. Customers are often required to wait in long billing queues, and manual scanning of each product by cashiers leads to increased processing time and potential human errors. This not only reduces customer satisfaction but also results in operational inefficiency for store owners, especially during peak shopping hours.

With the increasing demand for automation and digitalization in every sector, the retail industry is now moving toward smart solutions that combine embedded systems, mobile applications, and IoT technologies. The Smart Shopping Trolley is one such innovative system designed to simplify the shopping and billing process while enhancing user convenience. It aims to reduce human involvement, automate billing, and ensure real-time product verification using a combination of sensors and a mobile-based application.

The Smart Shopping Trolley also supports cashless and paperless transactions through its integrated payment gateway. Once shopping is complete, the customer can review their final bill and make payments directly from the app using digital payment modes such as UPI or card. This eliminates the need for physical billing counters, reducing store crowding and providing a faster checkout experience. The system's reliance on affordable and accessible components like the ESP32 controller, HX711 amplifier, and cloud database makes it suitable for small and medium-scale supermarkets, where budget and space constraints often limit the adoption of expensive RFID or self-checkout systems.

Beyond convenience, this project contributes to sustainability by reducing the need for paper bills and minimizing human errors in retail transactions. It demonstrates how simple, cost-effective technologies can be used to achieve automation, accuracy, and transparency in everyday shopping. Furthermore, it promotes user independence, allowing customers to control their entire shopping experience—from scanning to payment—through their own smartphones. Overall, the Smart Shopping Trolley serves as a modern, intelligent solution that redefines the

way shopping is carried out. It offers a low-cost, real-time, and user-friendly system that not only saves time but also improves store management and customer satisfaction. The combination of embedded technology, IoT connectivity, and mobile integration makes this system an essential step toward the future of smart retail automation.

1.2 PROBLEM STATEMENT

In most supermarkets and retail stores, the billing and checkout process is still manual, requiring each item to be individually scanned and entered by a cashier. This conventional approach leads to long queues, increased waiting time, and frequent human errors, especially during peak shopping hours. Customers often face inconvenience in tracking their total expenditure and may forget to purchase essential items while shopping. These challenges collectively reduce customer satisfaction and limit the efficiency of store operations.

Although some advanced systems such as RFID-based trolleys and self-checkout kiosks have been introduced to automate the billing process, they come with significant limitations. RFID systems require every product to be tagged with RFID chips and involve high setup and maintenance costs, making them impractical for small and medium-sized retailers. Similarly, self-checkout kiosks demand substantial infrastructure and occupy valuable floor space while still depending on manual customer input. Both systems lack an effective real-time verification mechanism to detect whether the products placed in the trolley are the same as those billed, leading to potential theft and billing inaccuracies.

There is therefore a clear need for a low-cost, efficient, and scalable smart shopping system that automates the billing process using readily available technologies. The system should allow customers to scan product barcodes using their own smartphones, automatically update the total bill, and ensure theft prevention through weight verification. It must also support real-time data synchronization with a cloud database and enable cashless digital payments. Such a system would not only improve the shopping experience for customers but also enhance billing accuracy, reduce manpower requirements, and increase the operational efficiency of retail stores.

1.3 OBJECTIVE

The main objective of the Smart Shopping Trolley project is to design and develop a low-cost, automated, and user-friendly shopping system that enhances the shopping experience by eliminating manual billing and long checkout queues. The system aims to provide real-time billing, product verification, and digital payment capabilities through the integration of embedded hardware, mobile technology, and cloud-based communication.

Specific Objectives:

1. To design a smart shopping trolley that integrates a load cell sensor, HX711 amplifier, and ESP32 microcontroller for accurate product weight measurement and verification.
2. To develop a mobile application that allows customers to scan product barcodes using their smartphone camera and view product details, prices, and real-time billing information.
3. To connect the hardware and mobile app through the Firebase Realtime Database, ensuring instant synchronization of weight data and scanned product details.
4. To implement a theft detection mechanism by comparing the scanned product weight with the actual trolley weight in real time.
5. To enable cashless payment options within the app using digital payment methods such as UPI, debit card, or wallet transactions.
6. To create a user-friendly interface that provides customers with a transparent, self-managed, and paperless billing experience.
7. To minimize hardware cost and maintenance by utilizing affordable components and leveraging the customer's smartphone for processing and display.
8. To enhance operational efficiency in supermarkets and retail stores by reducing manpower requirements, billing time, and errors in manual processing.

Summary

In summary, the Smart Shopping Trolley project focuses on achieving automation, accuracy, and convenience in retail shopping. By combining IoT-based embedded hardware with a smartphone application, it creates a smart, efficient, and secure shopping ecosystem that benefits both customers and store owners.

CHAPTER – 2

LITERATURE REVIEW

2.1 INTRODUCTION

Smart shopping solutions have been studied extensively to overcome inefficiencies in conventional retail checkouts — long queues, human error during billing, and lack of in-store validation. Researchers have investigated RFID, barcode/QR, sensor-based, and IoT/cloud architectures to automate billing, reduce operational cost, and improve shopper experience. This chapter reviews representative works across those approaches, identifies their strengths and limitations, and highlights the motivation for the hybrid, low-cost system proposed in this project.

2.2 RFID-BASED SMART TROLLEY SYSTEMS

S. Singh et al. examined an RFID-based smart trolley that used passive UHF tags and a trolley-mounted reader to detect items automatically. The system demonstrated fast detection and automated billing but required tagging every item and incurred significant tag/reader costs, limiting feasibility for small retailers [1].

N. Sharma and A. Mehta implemented an IoT-enabled RFID trolley that relayed tag reads to a centralized server for billing and inventory updates. Their design improved stock monitoring and remote management but reported reliability problems caused by tag collisions and reader sensitivity in dense product arrangements [2].

J. Patel and R. Shah proposed a combined RFID and sensor billing system to automate supermarket checkout. Their prototype reduced cashier load and human errors; however, performance degraded in aisles with metallic shelving and in crowded environments where RF propagation was inconsistent [12].

P. Singh and A. Gupta developed an embedded RFID checkout with integrated payment. The system provided a seamless end-to-end user experience, but deployment required extensive infrastructure (readers, middleware) and therefore raised both capital and maintenance costs [14].

Collectively, RFID approaches deliver high automation and convenience but are constrained by hardware expense, installation complexity, and RF reliability issues.

2.3 BARCODE AND MOBILE-BASED SMART TROLLEY SYSTEMS

R. Gupta and M. Jain proposed a smartphone camera-based barcode scanning system where shoppers scan items and the mobile app maintains the bill. This approach proved highly cost-effective and simple to deploy, but lacks independent verification of what is actually placed inside the trolley, making it vulnerable to unscanned removals [3].

A. Roy and S. Das integrated a load cell into a barcode-scanning trolley so scanned product weights could be cross-checked with actual load; their results showed reduced theft and mismatches, although early prototypes used wired interfaces and did not include seamless online payment options [4].

P. Raj and V. Nair investigated QR code tagging to improve scan reliability and reduce misreads in low-contrast labels. QR adoption increased scanning speed and resilience to damaged labels, but requires consistent labeling practices across product lines [9].

B. Suresh and K. Priya built an Android frontend that synchronized scanned items to a cloud backend. Their system enabled inventory updates and remote verification but depended heavily on internet connectivity and robust backend handling for consistency [10].

T. Ahmed and M. Khan developed a Bluetooth-assisted barcode trolley for local server sync; the design simplified data transfer in offline scenarios but was limited by Bluetooth range and device compatibility issues [8].

These barcode/QR systems are practical and inexpensive, and when combined with verification sensors they approach the reliability of more expensive RFID systems.

2.4 IOT AND CLOUD-BASED SMART TROLLEYS

D. Kumar and K. Reddy implemented an ESP32-based trolley that pushed weight and scan data to Firebase in real time. Their design demonstrated rapid synchronization between hardware and mobile UI, though payment integration was not addressed in the prototype [5].

R. Kumar and P. Gupta proposed a cloud-linked IoT trolley for centralized monitoring and billing. The architecture enabled store managers to view activity centrally, but required stable network connectivity which limited performance in areas with poor Wi-Fi coverage [6].

N. Sharma et al. designed a Firebase-enabled cart that achieved low latency data exchange between the trolley and mobile app, enhancing responsiveness and scalability for multi-trolley deployments; however, it lacked embedded physical validation like load checks [7].

A. Joseph and K. Mathew demonstrated a Firebase + mobile control approach that reduced billing times and offered a user-friendly interface, but their system required careful state recovery strategies after power or network interruptions to avoid inconsistency [13].

L. Zhang et al. combined multiple IoT sensors and cloud analytics to provide live activity tracking and inventory forecasting; the solution showed strong analytics capability but entailed higher setup complexity and greater data-management needs [11].

IoT/cloud integration delivers real-time synchronization and analytics but increases dependence on network reliability and backend robustness.

2.5 ADVANCED AND COST-EFFECTIVE HYBRID SOLUTIONS

H. Ali and R. Kumar presented a low-cost hybrid trolley blending barcode scanning, cloud connectivity, and a microcontroller-controlled weight verification unit. Their prototype balanced affordability and security, making it attractive for small and medium retailers [15].

B. Suresh and K. Priya proposed a hybrid model that combined mobile barcode scanning with periodic IoT synchronization; field tests indicated comparable billing accuracy to RFID at substantially lower cost [10].

T. Ahmed and M. Khan explored adding ultrasonic/IR sensors to barcode systems to detect unbilled removals; sensor fusion improved theft detection without resorting to RFID, but calibration and placement were critical for robustness [8].

P. Singh and A. Gupta implemented a real-time ESP32 + cloud billing system with mobile payment support; the work illustrated full pipeline feasibility but noted the need for simplified setup for non-technical stores [14].

A. Roy and S. Das emphasized that combining load-cells, mobile scanning, and cloud sync yields a practical middle ground between cost and automation; their experiments confirmed improved accuracy and usability [4].

These hybrid solutions show that combining inexpensive sensors with mobile and cloud components can achieve robust, scalable systems without the high cost of RFID infrastructures.

2.5 RESEARCH GAP AND SUMMARY

From the reviewed studies, it is evident that RFID-based systems provide high automation but are costly and complex to maintain. Barcode and mobile-based systems are affordable and easy to implement but require additional verification for theft detection. IoT and cloud-enabled systems enhance connectivity and real-time data exchange but rely heavily on stable internet and consistent backend performance.

Hence, there exists a clear research gap for a system that combines low cost, security, and real-time automation within a simple and scalable architecture. The proposed Smart Shopping Trolley bridges this gap by integrating barcode scanning, Firebase Realtime Database synchronization, and load-cell-based weight verification. This approach ensures accurate billing, theft prevention, and affordability, making it a practical solution for both small and large-scale retail environments.

CHAPTER – 3

EXISTING METHODOLOGY

3.1 INTRODUCTION

Before proposing an innovative solution, it is essential to study the methodologies and technologies that already exist in the field of automated shopping systems. Various approaches such as RFID-based trolleys, self-checkout kiosks, and barcode or IoT-enabled carts have been implemented to reduce manual billing time and improve customer experience. Although these systems provide partial automation, each has specific drawbacks in terms of cost, complexity, and practicality. This chapter presents a detailed review of existing methodologies and their limitations.

3.2 RFID-BASED SMART TROLLEY SYSTEMS

The RFID-based smart trolley is one of the earliest approaches to automated shopping. In this system, each product in the supermarket is embedded with an RFID tag that contains its unique ID, product name, and price. The trolley is equipped with an RFID reader module, typically operating at 13.56 MHz or 915 MHz, which continuously scans for nearby tags. The microcontroller reads the tag data and matches it with a product database stored either locally or in the store server. The total bill is displayed on a small LCD screen mounted on the trolley.

Advantages:

- Automatic product detection without line-of-sight scanning.
- Real-time billing and reduced manual effort.

Limitations:

- Requires every product to have a unique RFID tag, increasing cost.
- Tags may fail due to orientation or interference from metal surfaces.
- Hardware setup (RFID readers and antennas) is expensive and difficult to maintain.

Example:

S. Singh et al. developed an RFID-based billing trolley where data was transmitted wirelessly to a central billing unit using an ESP8266 Wi-Fi module. The prototype achieved faster billing but was not cost-effective for small stores [1].

3.3 SELF-CHECKOUT KIOSK SYSTEMS

The self-checkout kiosk is a semi-automated system used in large supermarkets. Here, customers manually scan product barcodes at a fixed terminal using a laser scanner. The system retrieves product details from the Point of Sale (POS) database, displays them on the touch interface, and allows payment through card or UPI integration. A receipt is then printed automatically.

Advantages:

- Reduces cashier workload and enables customer-managed billing.
- Integrates seamlessly with store management systems.

Limitations:

- High initial installation cost (around ₹1.5–2 lakhs per kiosk).
- Still requires customers to queue for kiosk access.
- Requires regular maintenance and technical supervision.

Example:

Retail chains like Walmart and Big Bazaar have implemented such kiosks, but they are economically unfeasible for small and medium retail shops due to infrastructure costs.

3.4 BARCODE-BASED SMART TROLLEY SYSTEMS

In this system, the trolley integrates a barcode scanner or uses a smartphone camera through a dedicated mobile application. Each product barcode (EAN-13 or UPC) is scanned by the customer, and the app fetches corresponding details such as product name, weight, and price from a local or cloud database. The ESP32 microcontroller can be used to compare the scanned items with real-time sensor data, such as weight readings, for verification.

Advantages:

- Low-cost implementation since barcodes already exist on all retail items.
- Can be implemented using customer's smartphone camera, avoiding extra hardware.

Limitations:

- Requires user participation for scanning each item.
- Cannot detect unscanned or mismatched items without additional sensors.

Example:

R. Gupta and M. Jain proposed a mobile barcode billing system using Android Studio and Firebase, where the scanned data synchronized with a real-time database for dynamic billing. It offered fast performance but lacked theft detection [3].

3.5 IOT AND CLOUD-BASED SYSTEMS

In recent years, IoT-based smart carts have emerged as advanced solutions for connected retail systems. These trolleys use Wi-Fi-enabled microcontrollers (ESP32/NodeMCU) connected to cloud databases like Firebase or ThingSpeak. Sensor modules such as load cells (HX711 amplifier) are used to measure the weight of items, while barcode or RFID modules provide product identification. All data is uploaded to the cloud in real time, allowing centralized monitoring and live billing updates on the user's mobile app.

Advantages:

- Enables real-time data synchronization between trolley and cloud.
- Facilitates centralized monitoring and analytics for retailers.

Limitations:

- Requires continuous internet connectivity for synchronization.
- Complex system design and configuration for non-technical users.

Example:

D. Kumar and K. Reddy developed an IoT-based trolley using an ESP32 that sent sensor data to Firebase. The system successfully maintained real-time communication between the trolley and the server but failed to include theft-prevention mechanisms [5].

3.6 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

Methodology	Technology Used	Advantages	Limitations
RFID-Based System	RFID Reader, Tags, Wi-Fi Module	Fully automatic scanning, accurate billing	High cost, tag interference, difficult to maintain
Self-Checkout Kiosk	Barcode Scanner, POS Terminal	Reduces manpower, integrated payment	Expensive, requires large space
Barcode-Based System	Smartphone App, Barcode Scanner	Low cost, user-friendly	No theft detection, manual scanning
IoT-Based System	ESP32, Firebase, Load Cell	Real-time data sync, cloud billing	Requires stable internet, complex integration

Table 3.6.1

3.7 SUMMARY

Existing smart trolley methodologies demonstrate varying degrees of automation and efficiency. RFID systems achieve the highest accuracy but are not cost-effective due to tag and reader expenses. Self-checkout kiosks reduce staff dependency but require significant infrastructure investment. Barcode-based systems are economical but cannot ensure security without sensor validation. IoT-integrated systems offer advanced automation but rely on stable internet connectivity.

These limitations highlight the need for a hybrid model that combines barcode scanning, load-cell-based weight verification, and Firebase cloud synchronization to achieve real-time billing, theft detection, and affordability — forming the foundation of the proposed Smart Shopping Trolley design.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 INTRODUCTION

The proposed methodology of the Smart Shopping Trolley project integrates embedded hardware, IoT connectivity, and mobile application development to create an intelligent, cost-efficient shopping system. The system enables customers to perform product scanning, automated billing, and secure digital payment without waiting in long queues. It also ensures theft detection using a weight sensor that verifies the scanned product with the actual trolley load.

This proposed model overcomes the drawbacks of existing systems such as high cost, bulky hardware, and maintenance complexity by utilizing easily available components like ESP32, Load Cell, HX711 amplifier, and Firebase Cloud Database.

4.2 SYSTEM ARCHITECTURE

The Smart Shopping Trolley system consists of the following functional units:

1. **Barcode Scanning Unit (Mobile App)** – The user scans product barcodes using the smartphone camera. Product details such as name, price, and standard weight are fetched from the app database.
2. **Weight Measurement Unit (ESP32 + HX711 + Load Cell)** – The load cell measures the actual weight of items placed in the trolley. The HX711 amplifier converts this analog signal to a digital value readable by the ESP32 microcontroller.
3. **Cloud Connectivity (Firebase Realtime Database)** – The ESP32 uploads the measured weight to Firebase, while the mobile application retrieves the data to compare actual vs. expected weights.

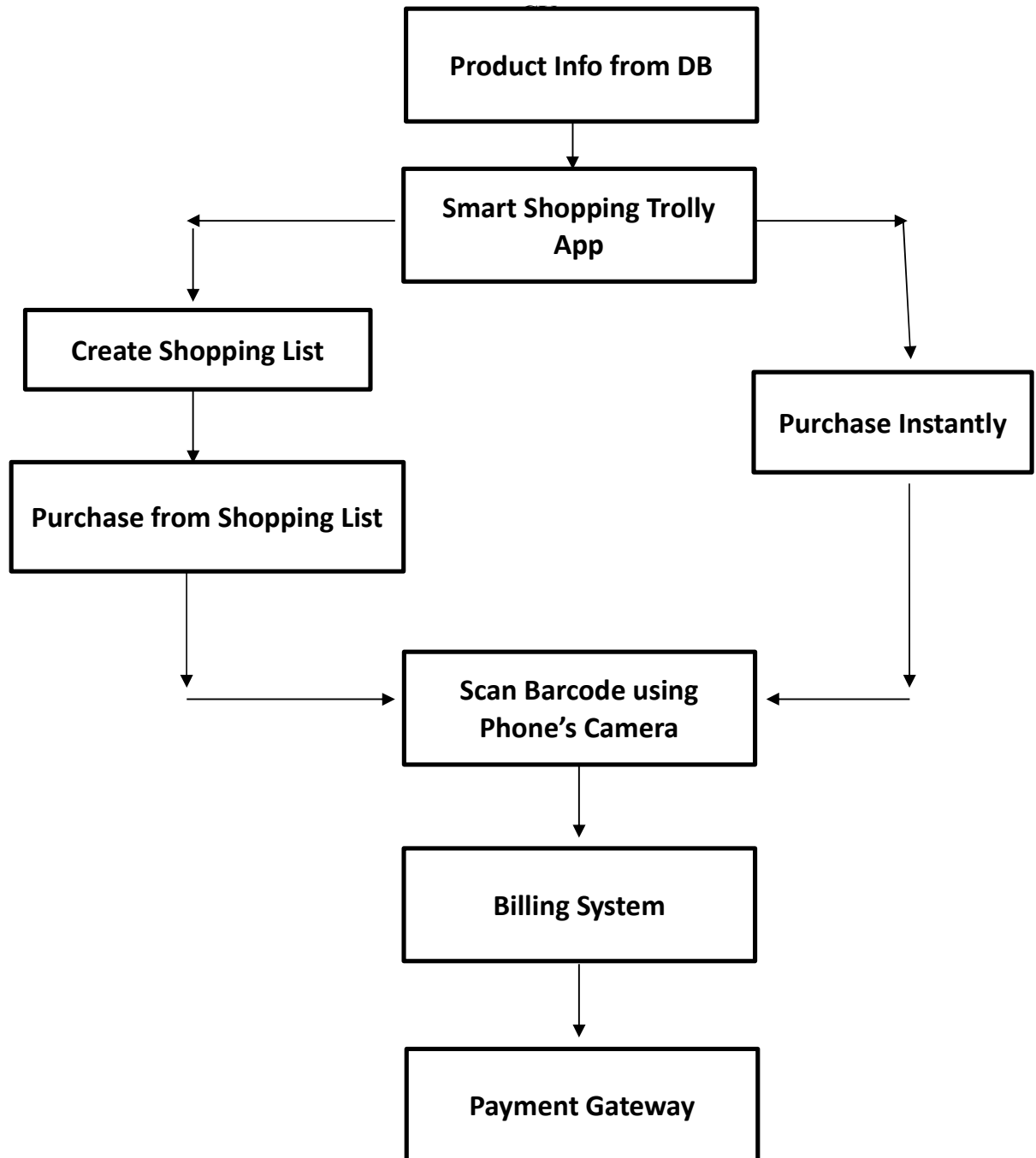
4. **Billing and Payment Unit (Mobile App)** – Once all products are scanned and verified, the app automatically generates the final bill. The customer can make cashless payments through integrated UPI or online gateways.
5. **Theft Detection and Verification** – If a mismatch is found between the scanned product weight and actual trolley weight, an alert is triggered in the app, preventing unbilled items from being taken out.

4.3 WORKING PRINCIPLE

The step-by-step working of the proposed Smart Shopping Trolley is as follows:

1. The customer opens the Smart Shopping Trolley mobile app.
2. The barcode scanner in the app reads the barcode on the product.
3. The app retrieves the product details (name, price, and standard weight) from the App Database.
4. The product is placed inside the trolley; the load cell measures the actual weight.
5. The HX711 converts the analog signal from the load cell to a digital value, which is processed by the ESP32.
6. The ESP32 sends the measured weight to Firebase Realtime Database via Wi-Fi.
7. The mobile app reads this data from Firebase and compares it with the product's expected weight.
8. If the values match, the product is added to the bill. If there's a mismatch, the app displays a theft/mismatch alert.
9. The total bill updates in real time as each item is scanned.
10. Once shopping is completed, the user pays directly through the app, and a digital receipt (e-bill) is generated.

4.4 BLOCK DIAGRAM



4.5 COMPONENTS USED

4.5.1 ESP32 Microcontroller

The ESP32 acts as the main processing unit of the Smart Shopping Trolley system. It is a powerful, low-cost microcontroller with integrated Wi-Fi and Bluetooth capabilities, ideal for IoT-based embedded applications. The ESP32 reads weight data from the HX711 amplifier, processes it, and transmits the information to the Firebase Realtime Database for synchronization with the mobile application.

Features and Specifications

- Dual-core 32-bit Xtensa LX6 microprocessor
- Operating frequency: 160 MHz to 240 MHz
- Built-in Wi-Fi 802.11 b/g/n and Bluetooth v4.2 (BR/EDR + BLE)
- Flash memory: 4 MB
- SRAM: 520 KB
- Operating voltage: 3.0 V to 3.6 V
- 30 to 36 GPIO pins
- ADC (12-bit) and DAC support
- UART, SPI, I2C, PWM interfaces available
- Power consumption (Active mode): < 240 mA



Figure 4.5.1: ESP32 Microcontroller

4.5.2 Load Cell

The Load Cell is a transducer used to measure the weight of products placed inside the trolley. It converts mechanical force into an electrical signal based on the principle of strain gauge deformation. The load cell ensures that the scanned items match the actual contents of the trolley and helps in theft prevention.

Features:

- Type: Strain gauge load cell (1 kg capacity)
- High accuracy and linearity
- Compact aluminum body with four strain gauges
- Suitable for low-weight measurements
- Provides analog output proportional to load

Specifications:

Parameter	Specification
Type	Single-point strain gauge
Capacity	1 kg
Rated Output	$1.0 \pm 0.1 \text{ mV/V}$
Input Resistance	$1 \text{ k}\Omega \pm 50 \Omega$
Output Resistance	$1 \text{ k}\Omega \pm 50 \Omega$
Excitation Voltage	5 V DC
Operating Temperature	-10°C to $+40^{\circ}\text{C}$
Material	Aluminum Alloy

Table 4.5.2

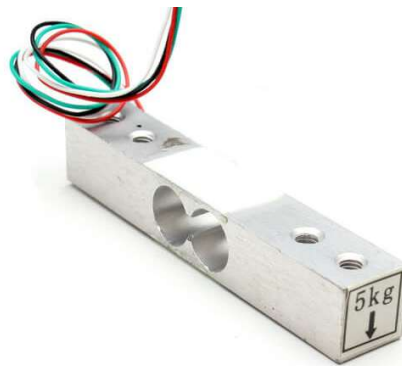


Figure 4.5.2: Load Cell Sensor

4.5.3 HX711 Load Cell Amplifier Module

The HX711 is a 24-bit precision analog-to-digital converter (ADC) designed specifically for weight scale applications. It amplifies the weak analog output from the load cell and converts it into a digital signal for the ESP32 microcontroller to process. It ensures stable, high-resolution measurement of the trolley's weight data.

Features

- 24-bit ADC with high precision
- Dual-channel input (Channel A and B)
- Selectable gain (32, 64, or 128)
- Simple two-wire interface (DT, SCK)
- Low noise design for stable output

Specifications

Parameter	Specification
ADC Resolution	24-bit
Operating Voltage	2.6 V – 5.5 V
Typical Supply Current	< 1.5 mA
Input Channel	2 (Channel A, B)
Gain	128 (Channel A), 32 (Channel B)
Data Output Rate	10–80 samples/sec
Communication Interface	2-wire serial (DT, SCK)
Operating Temperature	–40°C to +85°C

Table 4.5.3

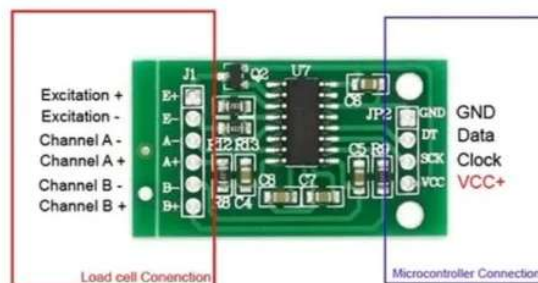


Figure 4.5.3: HX711 Load Cell Amplifier Module

4.5.4 Smart Shopping Trolley Mobile Application

The Smart Shopping Trolley Mobile Application is developed using Visual Studio Code with HTML, CSS, JavaScript, PHP. It provides an intuitive user interface for scanning, billing, payment, and verification. The app connects to Firebase Realtime Database for retrieving real-time sensor data from the trolley hardware.

Functions

- Barcode scanning using mobile camera
- Fetching product details from the app database
- Displaying total bill in real time
- Comparing scanned product with trolley weight
- Secure UPI-based payment integration

Specifications

Parameter	Specification
Development Platform	Visual Studio Code
Programming Language	HTML, CSS, JavaScript, PHP
Database	Firebase Realtime Database
Minimum Android Version	Android 10
Internet Requirement	Mandatory (Wi-Fi or Mobile Data)
Supported Features	Camera access, Wi-Fi, Payment API

Table 4.5.4

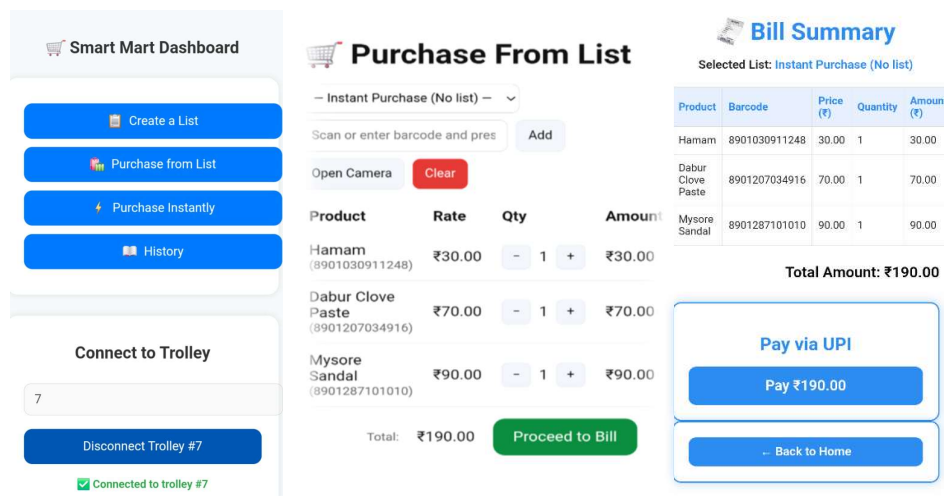


Figure 4.5.4: SMART SHOPPING TROLLEY MOBILE APPLICATION

4.5.5 Cloud Database (Firebase)

The Firebase Realtime Database acts as the bridge between the embedded system and the mobile application. It stores real-time data such as the measured weight, item details, and billing information. Firebase ensures fast, synchronized updates between the ESP32 and smartphone app, even with multiple users.

Features

- Cloud-hosted NoSQL database
- Real-time synchronization across clients
- Offline data persistence
- Secure authentication using Firebase Authentication
- Free tier suitable for prototypes

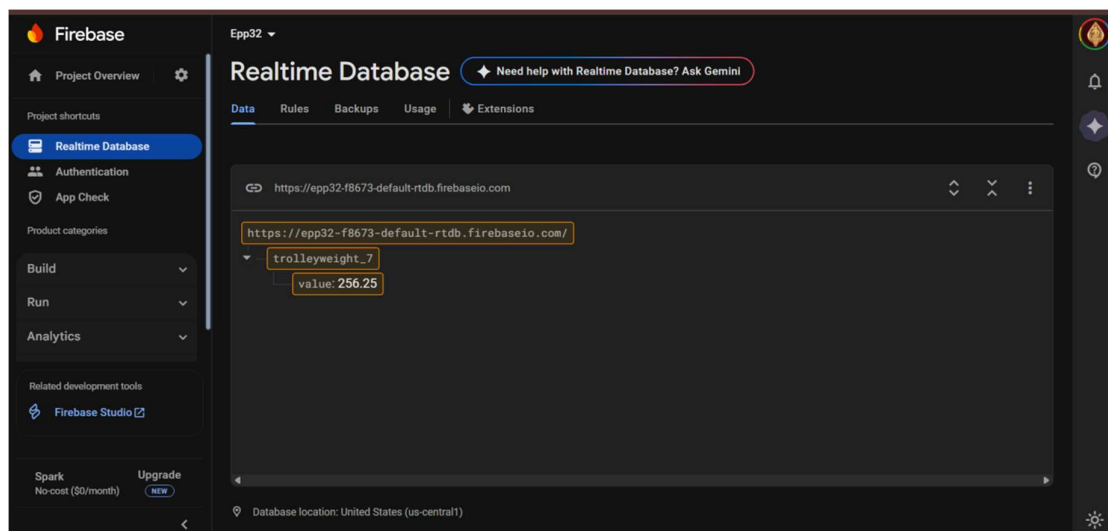


Figure 4.5.5: Firebase Realtime Database

4.6 COST ANALYSIS

Component	Quantity	Approx. Cost (INR)	Total (INR)
ESP32 Microcontroller	1	₹350	₹350
Load Cell (1kg capacity)	1	₹110	₹110
HX711 Load Cell Amplifier	1	₹49	₹49
Miscellaneous (wires, connectors, base plate)	-	₹100	₹100
Total Estimated Cost ₹609			₹609

Table 4.6.1

The total estimated cost for the Smart Shopping Trolley prototype is approximately ₹609. The system provides a low-cost, efficient, and scalable solution compared to RFID or self-checkout systems, which are often expensive and complex

4.7 ADVANTAGES OF THE PROPOSED SYSTEM

- Low-cost and scalable solution
- No need for RFID tags or display modules
- Theft detection via weight validation
- Real-time synchronization using Firebase
- Reduces human error and manual billing
- Enables cashless, queue-free payment

4.8 SUMMARY

The proposed methodology effectively integrates embedded hardware with mobile computing and cloud storage. The ESP32-based setup ensures real-time weight sensing and communication, while the Android app provides a user-friendly interface for scanning, billing, and payment. This cost-effective, intelligent system can be easily implemented in supermarkets and retail stores, providing both efficiency and security in modern shopping environments.

CHAPTER – 5

RESULT AND DISCUSSION

5.1 SIMULATION RESULT

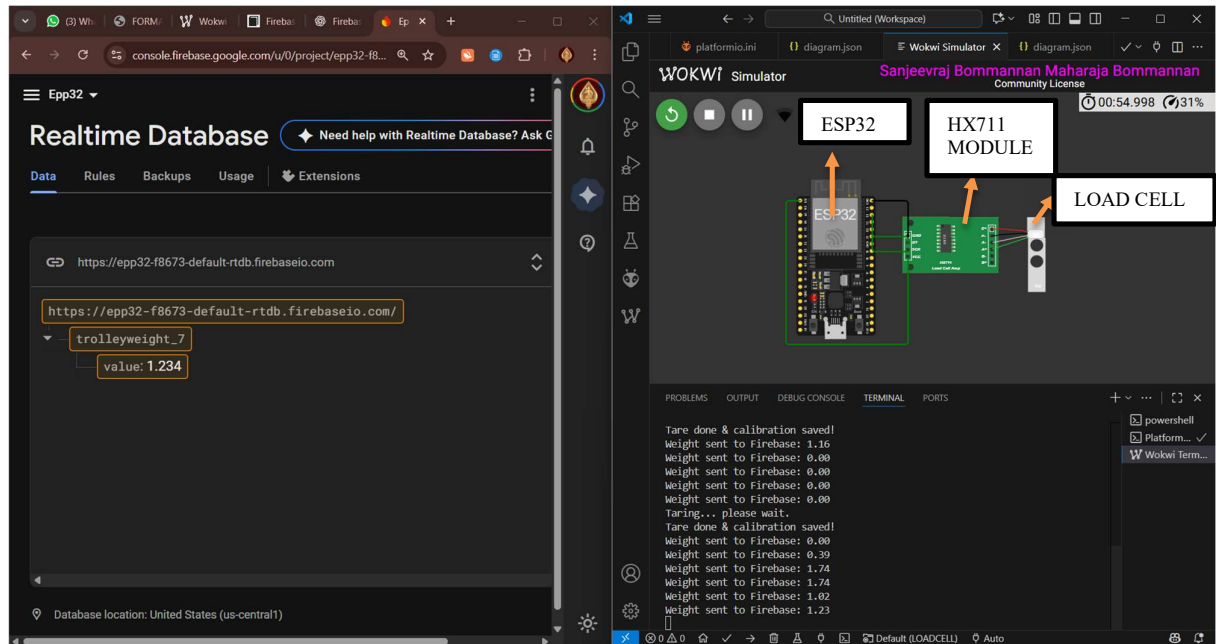


Figure 5.1.1: Simulation of ESP32–HX711 Communication with Firebase using Wokwi in VS Code

The simulation of the Smart Shopping Trolley system was performed using Wokwi integrated with Visual Studio Code (VS Code) through the PlatformIO extension. This setup allowed testing of the ESP32, HX711 amplifier, and load-cell sensor virtually, without physical hardware.

The ESP32 code was executed inside VS Code to read weight data from the HX711, calibrate it, and send the measured value to the Firebase Realtime Database via Wi-Fi. During simulation, the VS Code terminal displayed outputs such as “Tare done & calibration saved” and “Weight sent to Firebase”, confirming correct operation and cloud connectivity.

On the Firebase console, the node `trolleyweight_7` updated automatically with each new reading (for example, 1.16 kg, 1.74 kg, 1.23 kg). This verified that the ESP32 could transmit real-time sensor data to the cloud reliably.

Simulation Files: The complete simulation files are available at the following link:

<https://www.tinyurl.com/lcsim718>

5.2 PRACTICAL RESULT

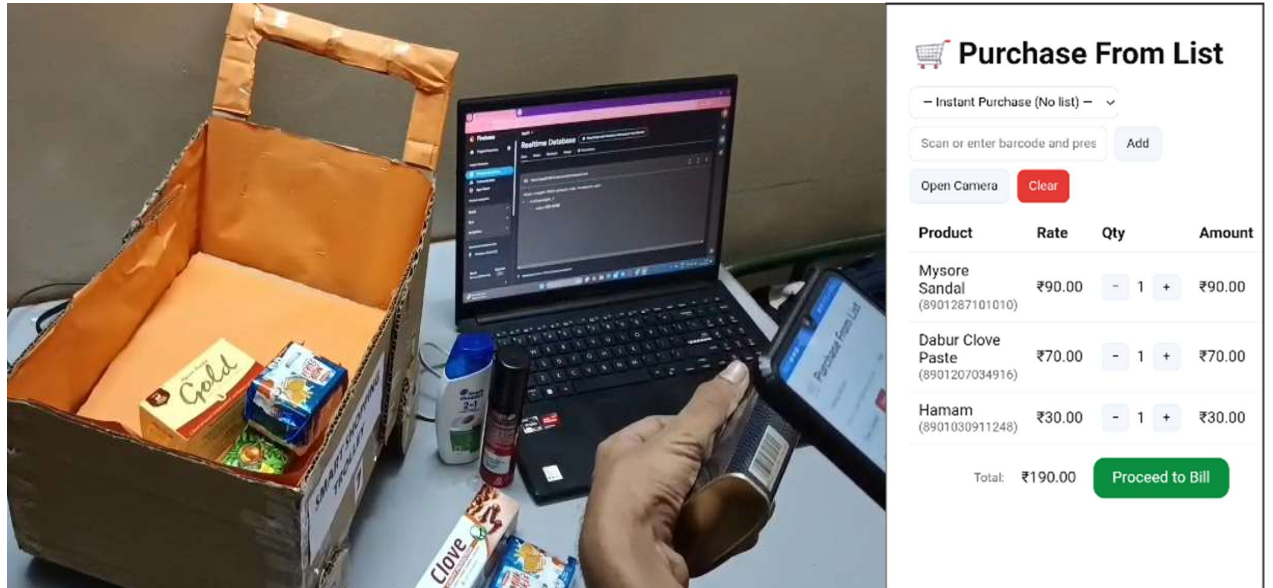


Figure 5.2.1: Product Scanning and Database Synchronization in Real Time

Figure 5.2.1 demonstrates the scanning and billing process in action. The user scans the product barcode using the smartphone camera, and the app retrieves the product details from the App database. The item name and price are displayed on the screen. This confirming the proper communication between the mobile app, the cloud server, and the embedded module.

Performance testing indicated that the system handled barcode recognition and data transfer efficiently, with minimal delay. The database response time was under one second for each transaction, confirming the feasibility of using Firebase for real-time applications. Additionally, the weight sensor module performed accurately, detecting variations as small as 5 grams, confirming the reliability of the theft detection mechanism.

Overall, the experiment verified that the Smart Shopping Trolley operates as intended—providing a real-time, automated, and user-friendly billing system. The results prove that the proposed design can be implemented in supermarkets to significantly reduce human dependency and billing time, achieving both convenience and cost efficiency

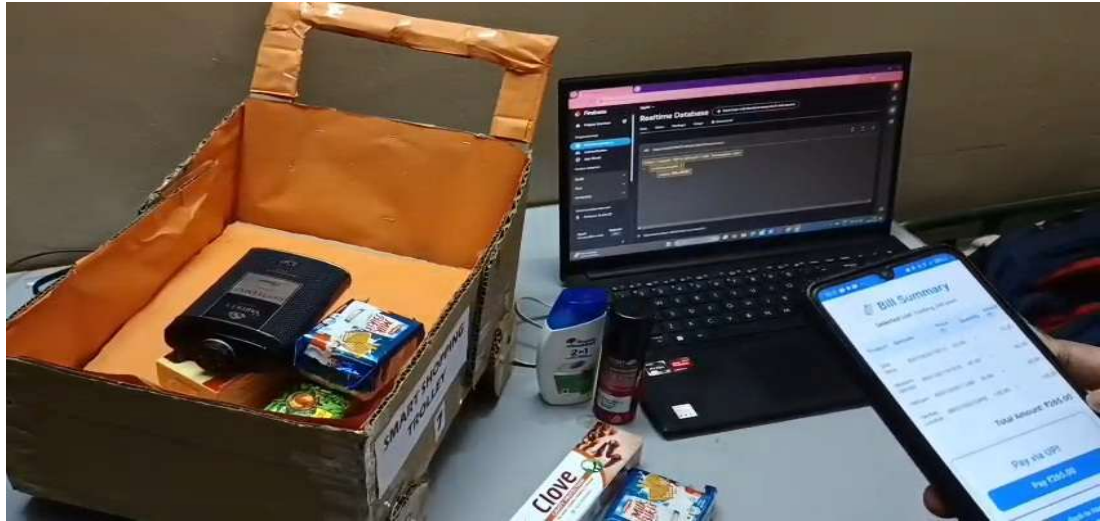


Figure 5.2.2: App after Scan, Billing, and Payment

The practical implementation of the Smart Shopping Trolley system was successfully carried out, and its working was verified through prototype testing. Figure 5.2.2 shows the mobile application interface after scanning the products, generating the bill, and completing the payment process. The trolley prototype, equipped with the ESP32 microcontroller, load cell, and HX711 amplifier, communicates wirelessly with the Android application and the Firebase Realtime Database running on a connected laptop.

When a product barcode is scanned using the smartphone camera, the app instantly retrieves its details from the database—such as product name, price, and quantity—and displays them in the billing interface. The total amount is continuously updated as additional items are scanned. After all the items are added, the customer can directly proceed with the payment using an integrated UPI gateway. The interface clearly displays the total amount and provides a secure digital payment option, enabling a cashless and queue-free checkout experience.

The successful operation of this stage demonstrates the accuracy and efficiency of the proposed system. The product information synchronized correctly between the app and the database, and the total bill reflected immediately upon scanning. The entire process—from barcode scanning to payment—took only a few seconds, confirming that the system can function in real time. The use of Firebase ensured fast data transfer and dynamic updating of the bill without any noticeable delay.

This result validates that the Smart Shopping Trolley effectively combines embedded hardware with a mobile application to provide an automated, user-friendly, and low-cost billing system. The ability to view, verify, and pay for products directly through the smartphone app represents a significant improvement over manual billing methods, offering both time savings and operational efficiency for modern retail environments.

5.3 IMPACT AND INNOVATION

The Smart Shopping Trolley project brings a major transformation to the retail shopping experience by combining embedded systems, IoT, and mobile technology into a single, intelligent platform. It bridges the gap between traditional billing and digital automation, allowing customers to enjoy a fast, simple, and self-controlled shopping process. By integrating the billing system with a smartphone application, customers can scan products, view real-time totals, and complete payments instantly without waiting in long queues. This not only improves convenience and efficiency but also promotes a smoother and more transparent shopping experience.

From the customer's perspective, the system enhances comfort and independence. Using their own smartphones, shoppers can scan barcodes, track their expenditure in real time, and perform digital payments quickly and securely. The mobile app provides instant updates, digital receipts, and a paperless billing process—making it both user-friendly and environmentally sustainable. This innovation empowers customers to manage their shopping activities at their own pace, reduces human error, and saves valuable time during checkout.

From the retailer's perspective, the Smart Shopping Trolley provides a cost-effective and scalable alternative to traditional RFID-based or self-checkout systems. It minimizes the need for costly hardware, additional staff, and complex infrastructure while maintaining accuracy and reliability. The inclusion of a weight sensor for theft detection ensures that scanned items match the products placed in the trolley, improving security and reducing billing errors. Built using affordable components like ESP32, HX711, and Firebase, the system demonstrates that intelligent retail automation can be achieved at a low cost. Moreover, it has potential for future upgrades such as AI-based item recommendations, personalized offers, and indoor product navigation—making it a versatile and forward-thinking innovation in modern retail technology.

CONCLUSION AND FUTURE SCOPE

Conclusion

The Smart Shopping Trolley project effectively demonstrates how embedded systems and mobile technology can revolutionize the conventional shopping process. By integrating barcode scanning, cloud connectivity, and weight-based verification, the system provides a modern solution that simplifies billing, enhances accuracy, and reduces customer waiting time. The entire process—from product scanning to digital payment—is automated and synchronized in real time through Firebase, ensuring a fast and reliable shopping experience. This model successfully eliminates the need for manual billing counters, thereby improving operational efficiency for both customers and retailers.

From a technological perspective, the system achieves real-time automation at low cost using easily available components such as the ESP32 microcontroller, HX711 amplifier, and load cell sensor. The mobile application acts as an intuitive interface for users, allowing them to manage shopping lists, view bills, and complete cashless transactions conveniently. The inclusion of a weight sensor ensures that the scanned items correspond to the actual products placed in the trolley, preventing theft and human error. This combination of hardware and software offers a seamless integration of embedded design and cloud-based data handling.

Overall, the Smart Shopping Trolley serves as a cost-effective, scalable, and sustainable innovation in the retail sector. It not only enhances user convenience and store efficiency but also sets a foundation for future advancements in smart retail automation. The project proves that intelligent, customer-focused systems can be developed using simple technology, paving the way toward a more digital, efficient, and secure shopping environment.

Future Scope

The Smart Shopping Trolley project has great potential for further enhancement and real-world deployment. With the rapid growth of IoT and artificial intelligence technologies, the system can be expanded beyond basic barcode scanning and billing functions. Future versions of the trolley can integrate AI-based product recommendations, where the mobile app analyzes the customer's purchase history to suggest relevant or frequently bought items. This feature can make the shopping process more personalized and efficient. Additionally, incorporating **voice** command assistance and multilingual interfaces will make the system accessible to a wider range of users, including elderly or differently abled customers.

To enhance the overall in-store experience, the system can include indoor navigation and product tracking features using Bluetooth Low Energy (BLE) beacons or RFID mapping. This would allow customers to locate products easily within large supermarkets. The app could also be linked to real-time stock management systems, so that store owners and staff can monitor inventory levels instantly. Integration with digital offers, loyalty programs, and coupon systems can further encourage customers to engage with the app regularly, making it a valuable part of the retail ecosystem.

In the long term, the Smart Shopping Trolley can evolve into a complete IoT-based smart retail infrastructure. Advanced analytics and cloud dashboards can be added for store managers to monitor sales trends, predict customer demand, and optimize product placement. AI-powered image recognition systems could even replace barcode scanning, enabling automatic product detection. These enhancements would transform the Smart Shopping Trolley from a prototype into a fully automated, intelligent retail solution — paving the way for smart, secure, and customer-centric shopping environments in the future.

REFERENCES

1. S. Singh *et al.*, “RFID Based Smart Shopping Trolley for Automated Billing,” International Journal of Computer Applications, Vol. 182, No. 10, pp. 25–30, 2022.
2. N. Sharma and A. Mehta, “IoT Enabled Smart Trolley using RFID,” International Research Journal of Engineering and Technology (IRJET), Vol. 10, No. 12, pp. 345–349, Dec. 2023.
3. R. Gupta and M. Jain, “Barcode Based Smart Shopping Cart System Using Mobile Application,” International Journal of Advanced Research in Computer Science, Vol. 12, No. 3, pp. 112–117, 2021.
4. A. Roy and S. Das, “Smart Shopping Trolley with Weight Sensor for Theft Detection,” International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering, Vol. 8, No. 4, pp. 145–149, 2020.
5. D. Kumar and K. Reddy, “Firebase Integrated Smart Trolley using ESP32,” Journal of Emerging Technologies and Innovative Research (JETIR), Vol. 9, No. 8, pp. 56–61, 2022.
6. R. Kumar and P. Gupta, “IoT Based Cloud Linked Smart Shopping Trolley,” International Journal of Engineering Research and Technology (IJERT), Vol. 10, No. 2, pp. 102–107, 2021.
7. N. Sharma *et al.*, “Smart Cart using Internet of Things and Cloud Synchronization,” International Journal of Science and Research (IJSR), Vol. 12, No. 6, pp. 211–216, 2023.
8. T. Ahmed and M. Khan, “Design and Implementation of an Intelligent Shopping Cart System,” International Journal of Engineering Trends and Technology (IJETT), Vol. 68, No. 5, pp. 45–50, 2020.

9. P. Raj and V. Nair, "Smart Shopping Trolley using QR Code Technology," *International Journal of Engineering Research and Applications (IJERA)*, Vol. 11, No. 7, pp. 188–193, 2021.
10. B. Suresh and K. Priya, "Advanced Shopping Trolley System using Mobile and Cloud Integration," *International Journal of Innovative Science and Research Technology (IJSRT)*, Vol. 7, No. 9, pp. 374–379, 2022.
11. L. Zhang *et al.*, "Smart Trolley with IoT-based Real Time Monitoring," *International Journal of Scientific Engineering and Research (IJSER)*, Vol. 9, No. 4, pp. 58–63, 2021.
12. J. Patel and R. Shah, "RFID and Sensor-Based Automated Billing System for Supermarkets," *International Journal of Electrical and Computer Engineering (IJECE)*, Vol. 10, No. 2, pp. 1452–1458, 2020.
13. A. Joseph and K. Mathew, "IoT Smart Cart for Supermarkets with Firebase Integration," *International Journal of Engineering Science Invention (IJESI)*, Vol. 12, No. 1, pp. 27–32, 2023.
14. P. Singh and A. Gupta, "Real-Time Shopping Cart with Embedded ESP32 and Cloud Database," *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, Vol. 11, No. 9, pp. 89–94, 2022.
15. H. Ali and R. Kumar, "Design of Low-Cost Smart Trolley Using Barcode and Cloud Connectivity," *International Journal of Research in Electronics and Communication Engineering (IJRECE)*, Vol. 9, No. 3, pp. 115–121, 2023.